MATH 202 Your name: _____

Learning target A1, version 1

Use definite integrals to find the area between two curves. Decide whether to slice horizontally or vertically.

Consider the finite region bounded by the curves $x + y^2 = 20$ and $x - y = 0$.

(a) Sketch the curves, shade in the region, and draw a representative slice.
(Should your slice be horizontal or vertical?)

(b) Write down an approximate expression for dA, the area of your representative slice.

(c) Write down a definite integral that computes the total area of the finite region.

(d) (If you have time) What if it was $x^2 + y = 20$ and $x - y = 0$? How would your setup be different?

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Learning target A1, version 2

Use definite integrals to find the area between two curves. Decide whether to slice horizontally or vertically.

Consider the finite region bounded by the curves $y = 1 - x^2$ and $y = 2 - 2x^2$.

(a) Sketch the curves, shade in the region, and draw a representative slice.
(Should your slice be horizontal or vertical?)

(b) Write down an approximate expression for dA, the area of your representative slice.

(c) Write down a definite integral that computes the total area of the finite region.

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Learning target A1, version 3

Use definite integrals to find the area between two curves. Decide whether to slice horizontally or vertically.

Consider the finite region bounded by the x-axis, the curve $x = (2-y)^2 - 3$,
and the line $y = 3 - \frac{1}{2}(x+2)$.

(a) Sketch the curves, shade in the region, and draw a representative slice.
(Should your slice be horizontal or vertical?)

(b) Write down an approximate expression for dA, the area of your representative slice.

(c) Write down a definite integral that computes the total area of the finite region.

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Learning target A1, version 4

Use definite integrals to find the area between two curves. Decide whether to slice horizontally or vertically.

Consider the finite region bounded by $3x = y^2$ and $9y = x^2$.

(a) Sketch the curves (use Desmos!), shade in the region, and draw a representative slice. (Would you rather have a horizontal slice or a vertical slice?)

(b) Write down an approximate expression for dA, the area of your representative slice.

(c) Write down a definite integral that computes the total area of the finite region.